

Homework Assignment: DevOps Environment Setup (CSE636-Session 2)

Objective

By the end of this assignment, you will have a working local DevOps environment with the following tools installed and running:

1. GitHub CLI (or Git CLI)
2. Microsoft Visual Studio Code
3. Docker Desktop
4. Jenkins running inside a Docker container

Tasks

1. Install GitHub CLI

- Install GitHub CLI (gh) or Git CLI on your host computer.
- Authenticate with your GitHub account.
- Run the command:
gh --version
or
git --version

Evidence (screenshot): [Paste here]

Notes (1–2 sentences on what you did/issues faced): [Write here]

2. Install Visual Studio Code

- Download and install Visual Studio Code.
- Install the GitHub Pull Requests and Issues extension.

Evidence (screenshot of VS Code open): [Paste here]

Evidence (screenshot showing GitHub extension): [Paste here]

Notes (1–2 sentences on what you did/issues faced): [Write here]

3. Install Docker Desktop

- Download and install Docker Desktop.
- Run the following commands to verify installation:
 `docker --version`
 `docker run hello-world`

Evidence (screenshot of `docker --version`): [Paste here]

Evidence (screenshot of hello-world container run): [Paste here]

Notes (1–2 sentences on what you did/issues faced): [Write here]

4. Run Jenkins in Docker

- Pull the Jenkins LTS image:
 `docker pull jenkins/jenkins:lts`
- Run Jenkins container:
 `docker run -d -p 8080:8080 -p 50000:50000 --name jenkins jenkins/jenkins:lts`
- Open Jenkins in browser at `http://localhost:8080`
- Retrieve the initial admin password:
 `docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword`
- Complete the setup wizard.

Evidence (screenshot of Jenkins setup page in browser): [Paste here]

Evidence (screenshot of `docker ps` showing Jenkins running): [Paste here]

Notes (1–2 sentences on what you did/issues faced): [Write here]

Submission

Submit this completed document as your homework file (with screenshots and notes filled in).